

Final Project

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2026-06-09

https://github.com/CarloGrouper/ENVS-193DS_spring-2026_final.git

Setup

```
suppressPackageStartupMessages(library(tidyverse))
suppressPackageStartupMessages(library(janitor))
suppressPackageStartupMessages(library(dplyr))
suppressPackageStartupMessages(library(gt))
suppressPackageStartupMessages(library(rstatix))
suppressPackageStartupMessages(library(here))
suppressPackageStartupMessages(library(DHARMA))
suppressPackageStartupMessages(library(ggeffects))
suppressPackageStartupMessages(library(ggplot2))

nests <- read_csv(here("data", "nest_data_final.csv"), show_col_types = FALSE)
```

Problem 1

1a

Part 1: They likely used a logistic regression, since they tested whether distance to the water edge predicts the probability of American Avocet microhabitat use.

Part 2: They likely used a chi-square test of independence, since they tested for a relationship between two categorical variables: bird species and habitat type.

1b

Part 1: Distance to the water edge was associated to the probability of American Avocet microhabitat use, suggesting that avocets may prefer habitats at certain distances from the water's edge.

Part 2: Habitat use differed among American Avocets, Forster's Terns, and Black-necked Stilts, indicating that these species use habitat types in different ways.

1c

One additional variable could be native plant cover (%), measured as the percent of ground area covered by native vegetation within a $1m^2$ sampling quadrat. This is a continuous variable and may influence American Avocet habitat use because vegetation cover can affect food availability, visibility of predators, and nesting habitat suitability.

Problem 2

2a

```
nests_clean <- nests |>
  select(case_control, height_cm, ant_species) |>
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  mutate(
    ant_species = factor(
      ant_species,
      levels = c("AB", "RRB", "BBR")
    ),
    ant_full = case_when(
      ant_species == "AB" ~ "Crematogaster sjostedti",
      ant_species == "RRB" ~ "Crematogaster mimosae",
      ant_species == "BBR" ~ "Crematogaster nigriceps"
    )
  )

str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
```

```
$ height_cm : num [1:270] 392 310 513 154 324 ...
$ ant_species : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
$ ant_full : chr [1:270] "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaster mimosae"
```

```
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
  case_control height_cm ant_species ant_full
      <dbl>      <dbl> <fct>      <chr>
1         0      112   BBR      Crematogaster nigriceps
2         0      154.  RRB      Crematogaster mimosae
3         0      134.  RRB      Crematogaster mimosae
4         0      118   RRB      Crematogaster mimosae
5         0      327   RRB      Crematogaster mimosae
6         0      311   RRB      Crematogaster mimosae
7         0      146.  RRB      Crematogaster mimosae
8         0      114   RRB      Crematogaster mimosae
9         0       97.5 RRB      Crematogaster mimosae
10        0      316   RRB      Crematogaster mimosae
```

2b

A value of 1 means a tree had a bird nest present (nest occurrence), while 0 means the tree was an available/non-nest tree used as a control. Biologically, this represents whether or not birds chose that tree for nesting.

2c

Tree height is a continuous variable measured in centimeters. As tree height increases, the probability of nest occurrence may increase because taller trees may provide better visibility, predator avoidance, or structural support for nests. This information comes from the Methods section describing nest site selection variables and tree structural measurements.

Acacia ant species is a categorical variable with three levels (AB, RRB, BBR). It could influence nest occurrence because different ant species may vary in aggressiveness or protection of trees, which could either attract or deter birds from nesting. This is described in the Introduction and Methods sections discussing ant–bird interactions and nesting behavior.

2d

N/A

2e

```
model1 <- glm(case_control ~ height_cm + ant_species,  
              data = nests_clean,  
              family = binomial)  
  
model2 <- glm(case_control ~ height_cm * ant_species,  
              data = nests_clean,  
              family = binomial)
```

2f

```
AIC(model1, model2)
```

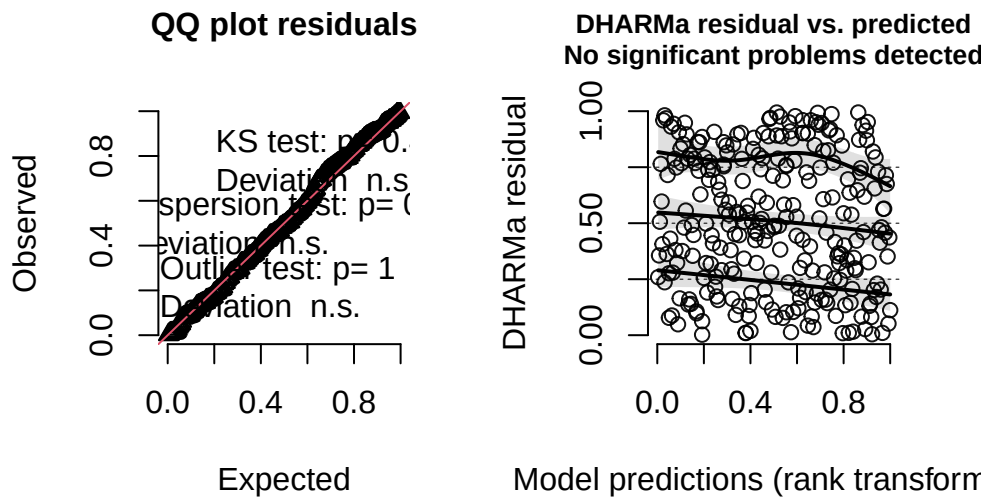
	df	AIC
model1	4	258.7430
model2	6	237.8042

The best model as determined by Akaike's Information Criterion (AIC) is the model that includes tree height, acacia ant species, and their interaction because it has the lower AIC value.

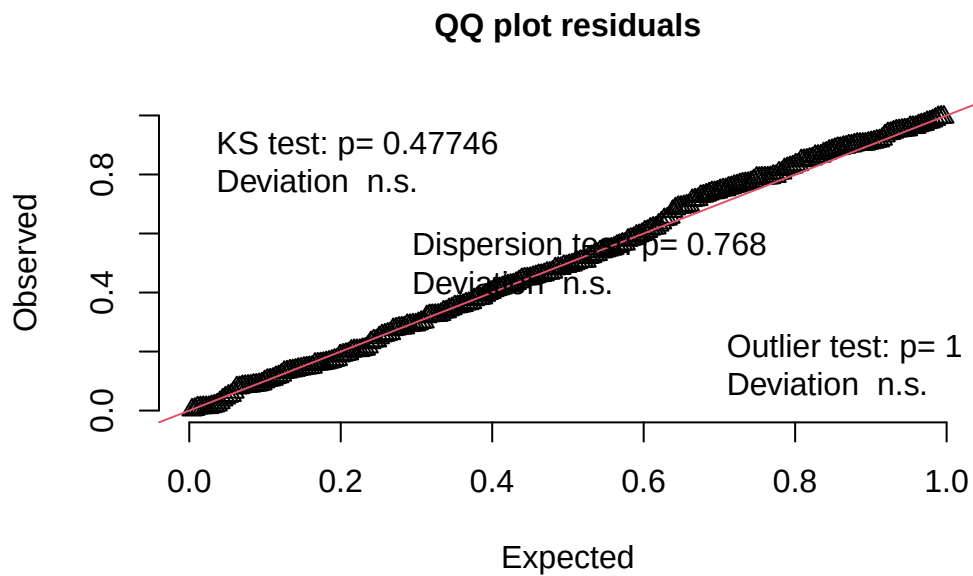
2g

```
sim_res <- simulateResiduals(fittedModel = model2)  
  
plot(sim_res)
```

DHARMA residual



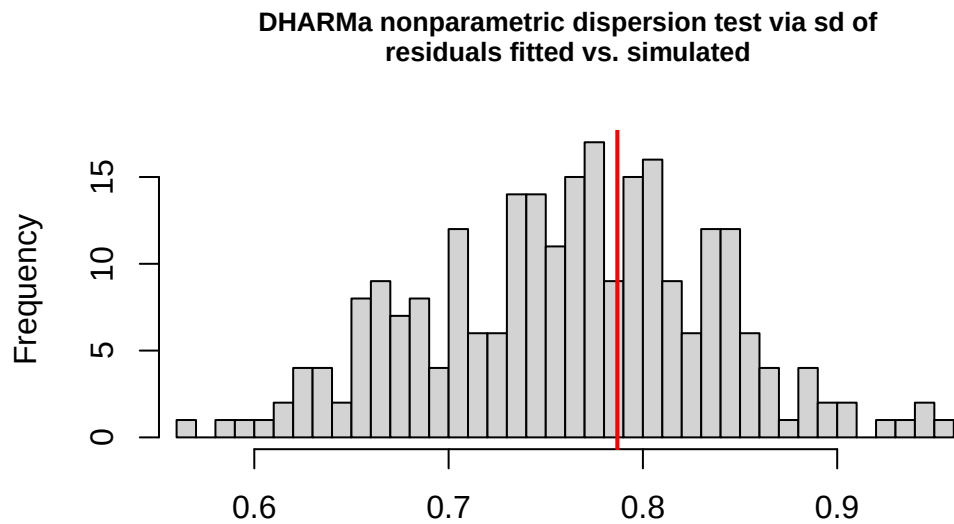
```
testUniformity(sim_res)
```



Asymptotic one-sample Kolmogorov-Smirnov test

```
data: simulationOutput$scaledResiduals  
D = 0.051246, p-value = 0.4775  
alternative hypothesis: two-sided
```

```
testDispersion(sim_res)
```



Simulated values, red line = fitted model. p-value (two.sided) = 0.768

DHARMA nonparametric dispersion test via sd of residuals fitted vs.
simulated

```
data: simulationOutput  
dispersion = 1.0325, p-value = 0.768  
alternative hypothesis: two.sided
```

2h

N/A

2i

N/A

2j

```
# Create a new data frame for prediction at 600 cm
new_data <- data.frame(
  height_cm = 600,
  ant_species = levels(nests_clean$ant_species)
)

# Get predicted probabilities
predict(model2, newdata = new_data, type = "response")
```

	1	2	3
	0.2048017	0.7407857	1.0000000

2k

At a tree height of 600 cm, the predicted probability of nest occurrence varies strongly by ant species, with relatively low probability for one species (approximately 0.20), moderate probability for another (approximately 0.74), and very high probability for the third (approximately 1.00). As shown in the figure, the relationship between tree height and nest occurrence differs across acacia ant species, indicating an interaction where the effect of height is not consistent across species. Overall, nest probability generally increases with tree height, but the strength of this increase depends on the ant species present. Biologically, this pattern likely reflects differences in how ant species influence nesting safety and suitability, such as variation in aggressiveness or protection of trees, which can make certain trees more attractive to nesting birds.

Problem 3

Setup

```
birds <- read_csv(here("data", "bird_weather_observations_copy.csv"), show_col_types = FALSE)
```

3a

In Homework 2, I used a boxplot (for weather type vs large bird sightings) and a scatterplot (temperature vs small bird sightings), while in Homework 3 I used a single scatterplot with both bird types shown together using different colors.

All visualizations are similar in that they represent bird observations collected over 20–30 minute periods across 35 days, and they all examine how environmental conditions (weather or temperature) relate to bird abundance. They also all use ggplot with point-based or distribution-based displays to show variation in bird counts over time.

Across all plots, there is an overall pattern suggesting that warmer or sunnier conditions are associated with more bird sightings, especially for larger birds. Homework 2 shows this more indirectly (through higher large bird counts under sunny conditions and a slight positive temperature relationship), while Homework 3 shows a clearer positive correlation between temperature and bird sightings for both small and large birds. The difference in clarity likely comes from Homework 3 having more complete data (35 rows) and combining both bird types into one visualization, making trends easier to compare directly.

I did not complete homework 3 at the correct time, so I did not receive feedback from my peers.

3b

A suitable analysis is to run two separate linear regression models predicting bird counts from temperature and environmental conditions. One model uses BigBird counts as the response variable, and the other uses SmallBird counts as the response variable, both with predictors such as temperature and atmospheric conditions.

This approach is appropriate because the response variables (BigBird and SmallBird) are continuous count variables, and the predictors include both a continuous variable (temperature) and a categorical variable (weather condition). Running separate models allows direct comparison of how environmental factors differently affect large and small birds across the 35 observation periods.

3c

Assumptions

```
bigbird_model <- glm(BigBird ~ Temp_F + Condition,
                     data = birds,
                     family = poisson)

smallbird_model <- glm(SmallBird ~ Temp_F + Condition,
                      data = birds,
                      family = poisson)

deviance(bigbird_model) / df.residual(bigbird_model)
```

```
[1] 0.7321749
```

```
deviance(smallbird_model) / df.residual(smallbird_model)
```

```
[1] 1.683467
```

The dispersion statistic for the BigBird model was 0.7321749, indicating no evidence of overdispersion. The dispersion statistic for the SmallBird model was 1.683467, which is somewhat greater than 1 and suggests mild overdispersion, but not enough to immediately reject the Poisson model.

Run the Analysis

```
summary(bigbird_model)
```

Call:

```
glm(formula = BigBird ~ Temp_F + Condition, family = poisson,
     data = birds)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.410716	1.092806	2.206	0.0274 *
Temp_F	-0.006986	0.017839	-0.392	0.6953
ConditionFair	0.227503	0.201152	1.131	0.2581
ConditionFog	-0.633192	0.526118	-1.204	0.2288
ConditionMist	-0.349042	0.244294	-1.429	0.1531
ConditionMosltyCloudy	0.045983	0.398904	0.115	0.9082

ConditionMostlyCloudy	0.526591	0.246556	2.136	0.0327 *
ConditionRain	0.472406	0.327759	1.441	0.1495
ConditionSmoke	-0.763237	0.410157	-1.861	0.0628 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 43.955 on 33 degrees of freedom
 Residual deviance: 18.304 on 25 degrees of freedom
 AIC: 167.66

Number of Fisher Scoring iterations: 4

```
summary(smallbird_model)
```

Call:

```
glm(formula = SmallBird ~ Temp_F + Condition, family = poisson,  
     data = birds)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.26128	0.99560	-0.262	0.79299
Temp_F	0.04651	0.01607	2.895	0.00380 **
ConditionFair	-0.11632	0.17427	-0.667	0.50449
ConditionFog	-0.55178	0.43078	-1.281	0.20024
ConditionMist	-0.26360	0.20364	-1.294	0.19550
ConditionMosltyCloudy	-0.45875	0.43865	-1.046	0.29564
ConditionMostlyCloudy	-0.11199	0.23827	-0.470	0.63835
ConditionRain	-1.29144	0.59127	-2.184	0.02895 *
ConditionSmoke	-0.98077	0.37748	-2.598	0.00937 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 79.336 on 33 degrees of freedom
 Residual deviance: 42.087 on 25 degrees of freedom
 AIC: 193.9

Number of Fisher Scoring iterations: 4

The number of large birds observed was not significantly related to temperature ($p = 0.6953$). Among weather conditions, only MostlyCloudy had a significant effect on large bird counts ($p = 0.0327$), with expected counts 1.6931503 times higher than the reference weather condition.

The number of small birds observed increased significantly with temperature ($p = 0.00380$). For every 1°F increase in temperature, the expected number of small birds increased by a factor of 1.0476133. Small bird counts were also significantly lower during Rain ($p = 0.02895$) and Smoke ($p = 0.00937$) conditions, with expected counts of 0.2748754 and 0.3750228 times those of the reference weather condition, respectively.

Effect size

```
exp(coef(bigbird_model))
```

(Intercept)	Temp_F	ConditionFair
11.1419307	0.9930381	1.2554616
ConditionFog	ConditionMist	ConditionMosltyCloudy
0.5308944	0.7053636	1.0470561
ConditionMostlyCloudy	ConditionRain	ConditionSmoke
1.6931503	1.6038490	0.4661553

```
exp(coef(smallbird_model))
```

(Intercept)	Temp_F	ConditionFair
0.7700660	1.0476133	0.8901927
ConditionFog	ConditionMist	ConditionMosltyCloudy
0.5759262	0.7682794	0.6320754
ConditionMostlyCloudy	ConditionRain	ConditionSmoke
0.8940548	0.2748754	0.3750228

For the BigBird model: MostlyCloudy weather increased expected large bird counts by 1.6931503 times relative to the reference condition.

For the SmallBird model: Each 1°F increase in temperature increased expected small bird counts by 1.0476133 times. Rain reduced expected small bird counts to 0.2748754 times the reference condition. Smoke reduced expected small bird counts to 0.3750228 times the reference condition.

3d

```
big_pred <- ggpredict(bigbird_model, terms = "Temp_F")
small_pred <- ggpredict(smallbird_model, terms = "Temp_F")

ggplot() +

  # Small bird data
  geom_point(
    data = birds,
    aes(x = Temp_F, y = SmallBird),
    color = "blue",
    size = 3,
    alpha = 0.7
  ) +

  # Big bird data
  geom_point(
    data = birds,
    aes(x = Temp_F, y = BigBird),
    color = "red",
    size = 3,
    alpha = 0.7
  ) +

  # Small bird prediction
  geom_line(
    data = small_pred,
    aes(x = x, y = predicted),
    color = "blue",
    linewidth = 1
  ) +

  # Big bird prediction
  geom_line(
    data = big_pred,
    aes(x = x, y = predicted),
    color = "red",
    linewidth = 1
  ) +

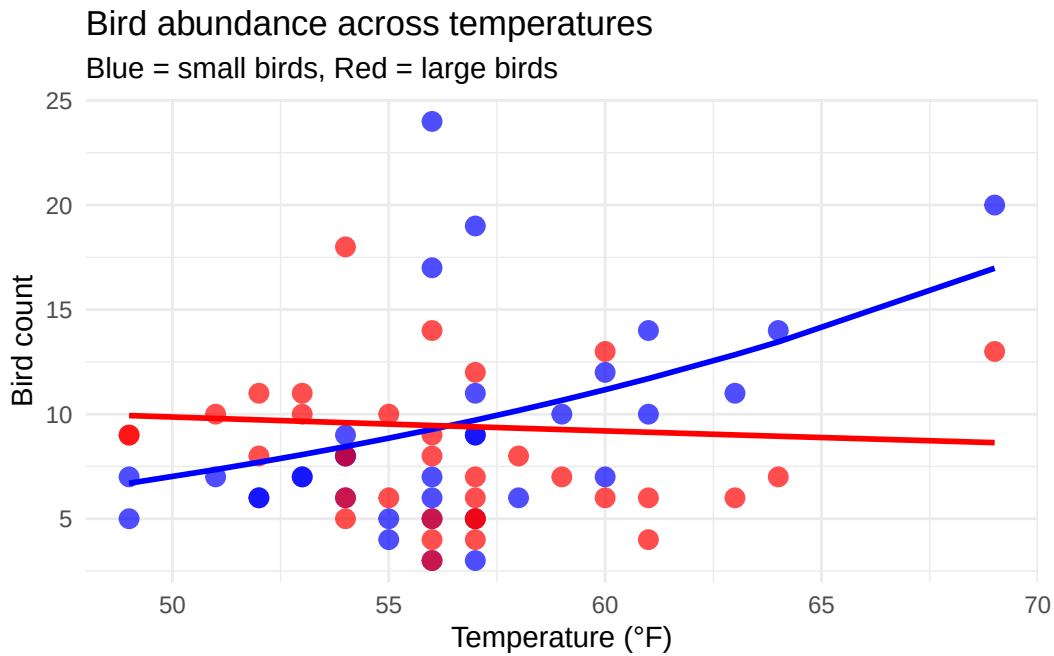
  labs(
```

```

title = "Bird abundance across temperatures",
subtitle = "Blue = small birds, Red = large birds",
x = "Temperature (°F)",
y = "Bird count"
) +

theme_minimal()

```



3e

Figure 1. Relationship between temperature and bird abundance. Blue points represent observations of small bird counts, while red points represent observations of large bird counts recorded during 20–30 minute observation periods. Blue and red lines show the predicted bird counts from Poisson regression models for small and large birds, respectively. The figure illustrates how bird abundance changes with temperature (°F), with the underlying data displayed as individual points.

3f

Small bird abundance increased significantly with temperature (Poisson regression: $\beta = 0.04651$, $z = 2.895$, $p = 0.00380$). The effect size indicates that each 1°F increase in temperature was

associated with a 1.0476133-fold increase in the expected number of small birds. In contrast, large bird abundance was not significantly related to temperature ($\beta = -0.006986$, $z = -0.392$, $p = 0.6953$). Small bird counts were also significantly lower during rain and smoke conditions compared to the reference weather condition (Rain: $p = 0.02895$; Smoke: $p = 0.00937$).

3g

I attended workshop on week 10 and did indeed complete my affective visualization.